

A stock price prediction method based on meta-learning and variational mode decomposition

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ABSTRACT

Stock price prediction is an important and challenging research topic, which has wide application prospects. Correct forecasting results can provide valuable guidance to investors and thus reduce the investment risk. To improve the prediction accuracy and obtain better prediction results, a new stock price prediction model called VML is proposed in this paper. First, the VML model slices the stock price series to obtain multiple window series, then uses variational mode decomposition (VMD) to decompose the window series to obtain multiple subseries. Unlike existing decomposition-based methods, VML decomposes the window series to solve the data leakage problem. Next, model-agnostic meta-learning (MAML) algorithm and long short-term memory (LSTM) network are applied to predict the subseries. A method of dividing the decomposed subseries into multiple tasks is proposed for the purpose of utilizing the MAML algorithm to train the initial parameters of the LSTM with good generalization ability. The initial parameters enable LSTM to fine tune dynamically to fit the latest data distribution of stock price data, which mitigates the impact of concept drift on prediction accuracy. Finally, the VML model merges the prediction results of the subseries to obtain the final predicted stock price. Experimental results on stock datasets of the Chinese Stock Market and the American Stock Market demonstrate that the proposed method improves the accuracy of prediction.

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1. Introduction

Correct stock price prediction results are helpful for investors to develop investment strategies and get favorable returns. However, stock prices are influenced by many factors such as policy, market, company's operation, and investors' behavior. Therefore, stock price series exhibit properties of non-linearity and high volatility, making stock price forecasting a challenging job.

As a classic problem in the intersection of finance and computer science, stock market prediction has attracted extensive attention from economists and computer scientists. Early stock price prediction methods such as Autoregressive Moving Average (ARMA), Autoregressive Integrated Moving Average (ARIMA), Autoregressive Conditional Heteroskedasticity (ARCH) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) were based on statistical linear prediction models. Linear models are more suitable for predicting smooth time series. Stock price series are usually nonlinear and non-smooth, so linear models do not perform well in the practical application of stock price prediction. With the development of machine learning, methods such as Support Vector Machine (SVM) [1], Random Forests [2],

and Hidden Markov Model (HMM) [3] have been used for stock price prediction. Compared with linear models, machine learning methods are more effective in handling nonlinear data and perform better in stock price prediction. In recent years, a number of deep learning methods have emerged, which can automatically learn complex feature representations from the non-smooth, nonlinear stock price data. Back Propagation Neural Network (BPNN) [4], Extreme Learning Machine (ELM) [5,6], Convolutional Neural Network (CNN) [7,8], Recurrent Neural Network (RNN) [9,10], Deep Belief Network (DBN) [11,12] and many other deep learning methods have been used for stock price prediction and have shown superior performance.

It is worth mentioning that RNNs are particularly suitable for time series prediction because of their capability to capture the correlation between previous and subsequent inputs. Long short-term memory network (LSTM) [13], a variation of RNN, alleviates the gradient vanishing problem of RNN and is widely used in stock price prediction [14–17]. Baek and Kim [14] proposed ModAugNet, an LSTM-based framework consisting of an over-fitting prevention module and a prediction module for forecasting stock market index. Liang et al. [15] proposed a new wavelet transform method for stock data preprocessing and then used LSTM to predict the preprocessed stock data. Ding and Qin [16] proposed a deep recurrent neural network model based on

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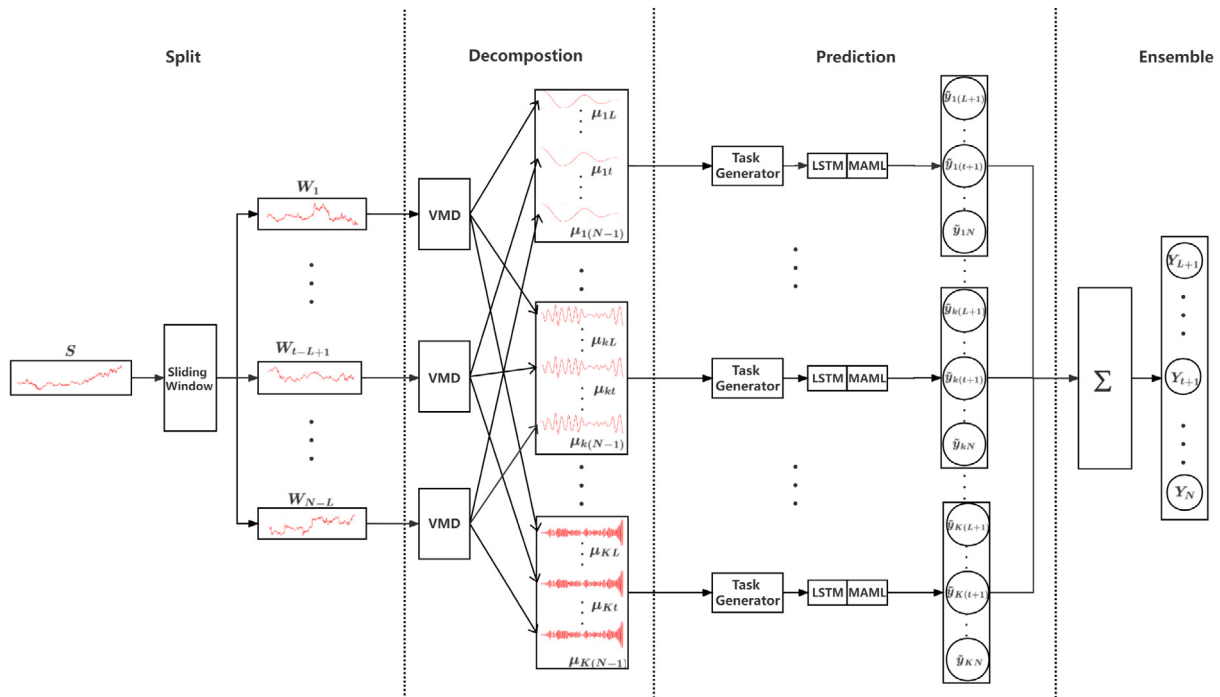


Fig. 1. The layout of the VML model.

LSTM with multiple inputs and outputs, which can simultaneously predict the opening, lowest and highest prices of stocks. Li et al. [17] proposed a method to extract dynamic time-delays to reconstruct multivariate data to improve the performance of attention-based long short-term memory (ALSTM) in industrial time series prediction.

Although neural networks show excellent predictive performance, there is no explicit mechanism to handle the non-stationarity of stock price series. To address this issue, decomposition-based hybrid models were proposed. The hybrid model combines the decomposition algorithms and forecasting models and decomposes the original time series into relatively smooth subseries for more efficiently constructing the forecasting models. Jothamani et al. [18] combined empirical mode decomposition and Artificial Neural Network (ANN) for short-term forecasting of Nifty stock index. Chen et al. [19] used empirical mode decomposition as the decomposition method, LSTM as the forecasting model, and introduced an attention mechanism for predicting the stock market index. Niu et al. [20] presented a hybrid model which combined variational mode decomposition and LSTM, improving the accuracy of the stock market index forecasting.

The decomposition-based hybrid models exhibit superior performance over some other models. Still, most existing decomposition-based hybrid models decompose the whole stock price series, including the training set and test set, therefore causing data leakage [21]. Data leakage will lead to look-ahead bias in the constructed model [22], which occurs when information or data used is not available during the analyzed period. If a model suffers from data leakage, the prediction results of the model will not provide valuable information for investors.

In addition, several factors influence stock prices, and the effects of these factors usually vary with time. Therefore, stock prices are dynamic, and their data distribution changes over time. That is, stock price series present a phenomenon of concept drift [23,24]. Most existing forecasting methods tend to construct a static model based on large amounts of historical stock price data. However, the performance of the static model is unstable

over time due to concept drift. If the model is trained with historical data and fine-tuned based on the latest data, the impact of concept drift can be weakened, thus improving the model's performance. Meta-learning attempts to tackle the "fast adaptation to new data" problem. The key idea of model-agnostic meta-learning (MAML) [25] is to train well-generalized initial parameters for the model on multiple tasks, hoping the model can adapt to new tasks through several gradient steps. However, different from the sample-based machine learning methods, meta-learning is task-based. It is a problem to divide the stock price data into different tasks when applying meta-learning to stock price prediction.

To solve the above problems, this paper constructs a new hybrid model (VML) for stock price prediction integrating meta-learning and decomposition-based model, as shown in Fig. 1. The model decomposes the stock price series with decomposition method and obtains a set of subseries. Compared with the original series, the subseries have stronger stability and regularity. But unlike the existing decomposition-based method, we first slice the stock price series with a sliding window, followed by decomposing the window series with VMD to obtain multiple subseries. Due to the decomposition of window subseries, the prediction results are independent of future information, eliminating the look-ahead bias. Then, the subseries are divided into multiple tasks to introduce the MAML [25] to train a widely generalizable initialization for LSTM. LSTM with the initialization will be fine-tuned on each task to fit its data distribution, thereby relieving the influence of concept drift. Finally, the prediction results of the subseries are summed to get the predicted stock price. We conduct extensive experiments on real-world stock data, and the experimental results show that the proposed method improves the accuracy of prediction. In summary, our major contributions are as follows:

- (1) We eliminate the look-ahead bias by changing the decomposition object from the whole series to the window series.
- (2) Meta-learning is introduced to alleviate the unstable performance of the model caused by the concept drift of stock price data. The problem of dividing stock price data into multiple tasks required by meta-learning is also solved.

(3) Integrating meta-learning and decomposition-based models, a new stock price prediction model VML is constructed, and experiments prove the model's effectiveness.

The remainder of this paper is organized as follows. Section 2 introduces the related work. Section 3 details the proposed method. Section 4 demonstrates the experiments on the stock price datasets. Section 5 concludes the work.

2. Related work

2.1. Decomposition-based hybrid methods

The decomposition-based hybrid method combines decomposition algorithm and prediction model for time series prediction. The main idea is “decomposition-prediction-ensemble”, that is, the original time series is decomposed into a set of subseries, then the prediction model is exploited to predict the subseries. The final prediction result is the ensemble of the results of these subseries. Though decomposition-based hybrid methods share the same idea, their structures are different. In Jothamani et al. [18], Chen et al. [19], Niu et al. [20], a model was used for each subseries to obtain prediction results of subseries respectively, and then merged the prediction results of subseries. In Zhou et al. [26], all the subseries were put into one prediction model together, and the final results were obtained directly from the model without further merging operation. While in Tang et al. [6], the original time series was first decomposed with variational mode decomposition (VMD) to obtain different modal components and the residual item, and then the residual item was decomposed by ensemble empirical mode decomposition (EEMD).

The decomposition-based hybrid method consists of two major components: the decomposition algorithm and the prediction model. Models such as ARIMA, GARCH, SVM, LSTM, and CNN can be treated as a prediction model of the hybrid method. As for the decomposition algorithms, the common ones are wavelet transform, empirical mode decomposition (EMD) [27], and variational mode decomposition (VMD) [28]. Fourier transform is one of the earliest decomposition methods proposed by Joseph Fourier in 1807 and is widely used in many fields [29–31]. However, Fourier transform has high requirements on the signal itself and does not work well for non-smooth and highly noisy signals. The wavelet transform decomposes the signal into the sums of more flexible functions, which is superior to Fourier transform. EMD is an adaptive signal processing method proposed by Huang et al. [27], which is especially suitable for the analysis and processing of nonlinear and non-stationary signals. EMD does not use any predetermined transform that depends on a specific theoretical structure [32,33], but is sensitive to noise and lacks mathematical theory. In order to improve EMD, Wu and Huang [34] proposed ensemble empirical mode decomposition (EEMD), a noise-assisted method. Dragomiretskiy and Zosso [28] proposed VMD, which was an entirely non-recursive model. VMD decomposed the original signal or sequence data into multiple components with a specific bandwidth in the spectral domain. Niu et al. [20], Jothamani and Başar [35] demonstrated that the VMD-based hybrid model had higher prediction accuracy than hybrid models based on other decomposition methods.

The subseries obtained by VMD is called Intrinsic Mode Function (IMF). The bandwidths of IMFs can be estimated by the following steps: (1) calculate the analytic signal correlating to each IMF by the Hilbert transform to obtain the unilateral frequency spectrum; (2) shift the frequency spectrum of each IMF to the corresponding “baseband”; (3) estimate the bandwidth by the

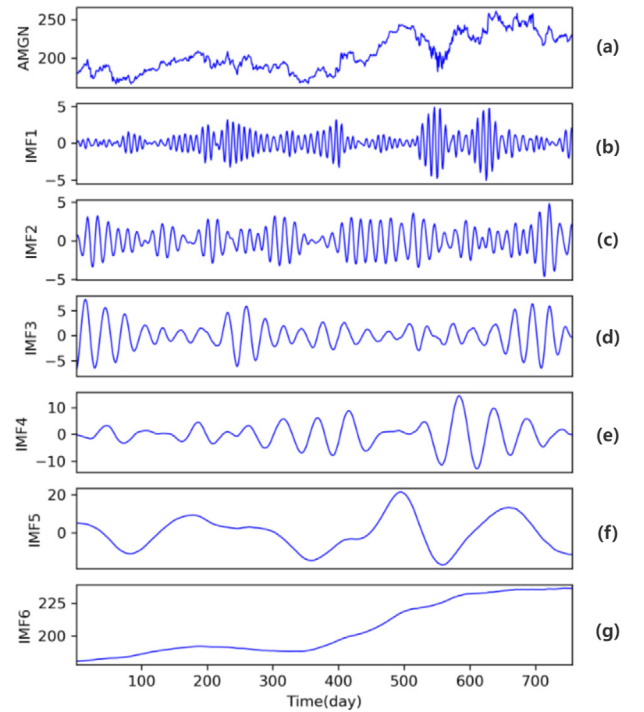


Fig. 2. The decomposition results of Amgen.

Gaussian smoothness of the demodulated signal. The constraints of the variational problem can be formulated as follows:

$$\min_{\{u_k\}, \{w_k\}} \sum_{k=1}^K \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-jw_k t} \right\|_2^2 \quad (1)$$

$$\text{s.t. } \sum_{k=1}^K u_k(t) = f(t)$$

where $u_k(t)$ denotes the IMF, $\{u_k\} = \{u_1, u_2, \dots, u_K\}$ denotes the set of IMFs, $\{w_k\}$ denotes the center frequencies of the u_k , $\{w_k\} = \{w_1, w_2, \dots, w_K\}$ denotes the set of center frequencies, $f(t)$ denotes the original signal, ∂_t denotes the differential processing of t , $\|\cdot\|$ indicates the norm processing, $\delta(t)$ is the Dirac function, and $*$ is the convolution symbol. The augmented Lagrange function is introduced to solve the optimization problem of constrained variational decomposition. The following form is obtained:

$$L(\{u_k\}, \{w_k\}, \lambda) = \alpha \sum_{k=1}^K \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-jw_k t} \right\|_2^2 \quad (2)$$

$$+ \left\| f(t) - \sum_{k=1}^K u_k(t) \right\|_2^2 + \left\langle \lambda(t), f(t) - \sum_{k=1}^K u_k(t) \right\rangle$$

where λ is the Lagrangian multipliers, and α is the quadratic penalty term. The above problem is solved using the alternate direction method of multipliers (ADMM) to obtain $\{u_k\}$, $\{w_k\}$. Fig. 2 shows the decomposition results of VMD, where Fig. 2(a) is the original stock price series and (b)–(g) are the subseries. As can be seen from the figure, subseries are more stable and regular than the original stock price series, so they can be modeled more effectively.

2.2. Meta-learning

Meta-learning, or learning to learn, was first proposed in the 1990s [36–38]. Meta-learning is task-based, and its goal is to improve the learning algorithm by learning on multiple tasks, hence making it quickly adapt to new tasks. A task T contains a task-specific dataset $D_T = \{D_S, D_Q\}$, where D_S denotes the support set for T , and D_Q is the query set for T . The mathematical expressions are as follows:

$$D_S = \{P_i = (s_i, l_i)\}_{i=1}^{N_S} \quad (3)$$

$$D_Q = \{P_i = (s_i, l_i)\}_{i=1}^{N_Q} \quad (4)$$

where $P_i = (s_i, l_i)$ denotes a sample pair, s_i is the sample, and l_i is the label corresponding to s_i . D_S consists of N_S sample pairs, and D_Q consists of N_Q sample pairs. There are two phases in meta-learning: meta-training and meta-testing. In the meta-training phase, the model is trained with a large number of tasks, which are called source tasks; in the meta-testing phase, the model performance is tested on new tasks, which are called target tasks.

Model-agnostic meta-learning (MAML) is one of the outstanding meta-learning algorithms, which is compatible with any model trained with gradient descent. In addition, MAML is the underlying framework of many meta-learning algorithms. It has been applied in various fields to deal with the traditional challenges in deep learning, such as data bottleneck and generalization. MAML was introduced to low-resource neural machine translation in [39]. Shi et al. [40] proposed a meta-learning framework based on the idea of MAML to predict market segment demand with limited data. Yao et al. [41] designed a spatial-temporal network with a meta-learning paradigm to tackle the problem of spatial temporal prediction for the cities with only a short period of data collection. By drawing on these studies, we frame stock price prediction as a meta-learning problem, and apply MAML to stock price prediction for the purpose of boosting the model's prediction performance.

3. Methodology

In this paper, a new stock price prediction model VML is constructed by combining meta-learning and the improved decomposition-based prediction model. VML model eliminates the look-ahead bias by decomposing the window series instead of the entire stock price series. At the same time, meta-learning is introduced for mitigating the impact of concept drift on prediction accuracy. The model consists of four modules, as shown in Fig. 1. The “Split” module slices the original stock price series S with a sliding window to obtain multiple window series $\{W_1, W_2, \dots, W_{N-L}\}$. The “Decomposition” module decomposes the window series $\{W_1, W_2, \dots, W_{N-L}\}$ with VMD to obtain more stable and regular subseries. Subseries $\{u_{kt}\}$ (Fig. 1) denotes the decomposition results of W_{t-L} . The stock price data is first divided into multiple tasks in the “Prediction” module. The good initialization θ for LSTM is trained with source tasks, and LSTM with initialization θ will fine-tune on each target task to adapt to the task. The prediction results of subseries $\{u_{kt}\}$ are $\{\hat{y}_{k(t+1)}\}$. The predicted stock price $Y_{(t+1)}$ at day $(t+1)$ is the ensemble of the prediction results $\{\hat{y}_{k(t+1)}\}$ of subseries $\{u_{kt}\}$, as shown in the “Ensemble” module.

3.1. Split and decomposition

Niu et al. [20] proposed VMD-LSTM for stock index forecasting, which is a hybrid model with VMD as a decomposition method and LSTM as a prediction model. The superior performance of VMD-LSTM has been shown experimentally. However,

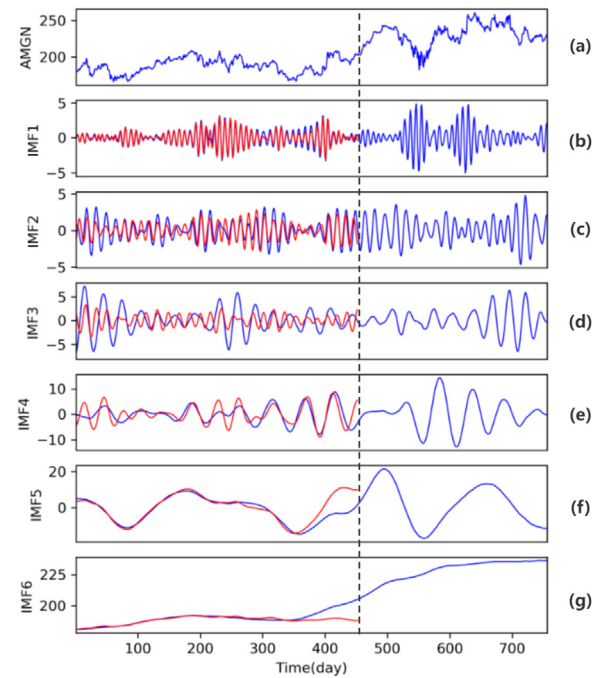


Fig. 3. The comparison between decomposition results.

VMD-LSTM decomposes the entire stock price series, including the training set and test set, resulting in a look-ahead bias [22]. Generally, the training set and test set are split in chronological order in stock price prediction. And the time area where training set and test set are located cannot intersect. Therefore, compared with the training set, the test set is the future information. However, VMD-LSTM first decomposes the whole stock price series and then splits the training set and test set. In the process of decomposition, the feature values involving the whole stock price series will be used, and knowledge of the hold-out test set will leak into the training set used to train the model, which will lead to the problem of data leakage.

To understand this, the data of Amgen (stock code: AMGN) from January 2, 2018 to December 31, 2020 is used to explain the data leakage problem of VMD-LSTM. The length of the whole stock price series is 756, and the length of the training set is 453. Fig. 3(a) is the original stock price series, and (b)–(g) are subseries obtained by VMD. Blue lines are decomposition results of the whole stock price series (756 days), and red lines are decomposition results of the training set (453 days). As shown in Fig. 3, the blue and red lines in (b)–(g) are not completely overlapping on the left side of the black dashed line ($x = 453$). The difference between the red line and blue line in (c) and (d) is particularly obvious. These results indicate that when the whole stock price series is decomposed first, and then the training set and test set are divided, knowledge of the test set will leak into the training set, resulting in data leakage.

If there is data leakage in the model, information not available at the time will be used, leading to look-ahead bias. Most existing decomposition-based stock price forecasting methods [18,19,26,42,43] similarly suffer from look-ahead bias. A back-testing simulation with look-ahead bias will not show accurate results. We decompose the window series instead of the entire stock price series to avoid the look-ahead bias. Specifically, the entire stock price series is first sliced into multiple window series with a sliding window. Then the window series will be decomposed to acquire the subseries. For example, given a stock price series $S = \{X_1, X_2, \dots, X_t, \dots, X_N\}$, where X_t is the stock price

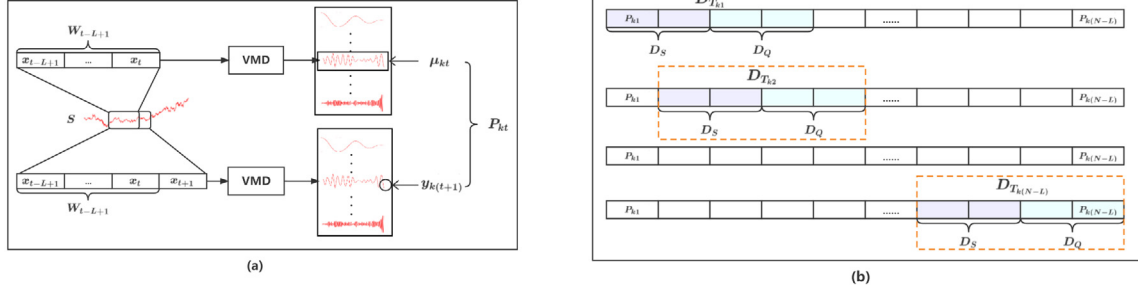


Fig. 4. Details of generating tasks.

on day t , $t \in [1, N]$, N is the length of the stock price series. S is sliced with a sliding window of size L to obtain multiple window series, as shown in the “Split” module of Fig. 1, where $W_i = \{X_i, X_{i+1}, \dots, X_{i+L-1}\}$ denotes the i th window series, $i \in [1, N - L]$. When predicting the stock price X_{t+1} on day $t + 1$, decompose the window series $W_{t-L+1} = \{X_{t-L+1}, X_{t-L+2}, \dots, X_t\}$ into K subseries $\{u_{kt}\} = \{u_{1t}, u_{2t}, \dots, u_{kt}\}$, $k \in [1, K]$ as shown in “Decomposition” module of Fig. 1. It is the window series W_{t-L+1} that is to be decomposed, which is historical information of X_{t+1} . Therefore, the subseries contain only historical information, which eliminates the data leakage problem and the look-ahead bias.

3.2. Prediction

MAML is introduced to stock price prediction with the aim of enabling the model to adapt quickly to new data. The problem of dividing stock price data into multiple tasks is the first to be addressed when applying MAML to stock price prediction. Then, stock prices can be predicted with the combination of MAML and LSTM. This section first discusses how to generate multiple tasks and then describes how to use MAML to train a well-generalized initialization for LSTM.

3.2.1. Generate tasks

As shown in Fig. 4(a), take the window W_{t-L+1} as an example to illustrate the formation of sample pairs. Firstly, the sequence $\{X_{t-L+1}, X_{t-L+2}, \dots, X_t\}$ in window W_{t-L+1} is decomposed to avoid the look-ahead bias. The obtained subseries $\{u_{kt}\}$ are taken as samples, that is, u_{kt} is the input of LSTM. Meanwhile, we decompose the sequence $\{X_{t-L+1}, X_{t-L+2}, \dots, X_t, X_{t+1}\}$ and take the last value $y_{k(t+1)}$ of the k th subseries as the label corresponding to the sample u_{kt} . Then, a sample pair $P_{kt} = (u_{kt}, y_{k(t+1)})$ is formed with the sample u_{kt} and the label $y_{k(t+1)}$. Next, D sample pairs are selected using a sliding window to constitute the data set $D_{T_{kj}}$ of task T_{kj} , $D_{T_{kj}} = \{P_{kj}, P_{k(j+1)} \dots P_{k(j+D-1)}\}$, $j \in [1, N - L]$, as shown in Fig. 4(b). In this way, the sample pairs contained in a task are continuous, and the time range of the sample pairs is not too long, so that the data distribution in a task will not change too much, and the impact of concept drift on the prediction results will be weakened. In each task, the first N_S sample pairs constitute the support set D_S , and the remaining N_Q sample pairs constitute the query set D_Q .

3.2.2. MAML

The idea of MAML [25] is to pre-train well-generalized initial parameters for the model on many source tasks, and the model with the initialization will adapt to the target task with only a few gradient steps. The stock price data are first split into the training set, validation set, and test set. Then, data of the three sets are decomposed by VMD and divided into multiple tasks separately.

The tasks obtained from the training set are used as source tasks to train the initialization of the model. And the tasks obtained from the validation set and the test set are used as target tasks to test the model's performance.

In the meta-training phase, a good initialization of the model is learned on source tasks. For a source task T , the model calculates the training error on the support set D_S and updates parameters with the following formula:

$$\theta^T = \theta - \alpha \nabla_{\theta} L(D_S^T, \theta) \quad (5)$$

$$L(D_S^T, \theta) = \sum_{(x,y) \in D_S^T} \|f_{\theta}(x) - y\|_2^2 \quad (6)$$

where θ are all parameters of the model, θ^T are the parameters of the model after updating on task T , α is the step size, and ∇_{θ} denotes the gradient of θ . $L(D_S^T, \theta)$ denotes the training error of the model with parameters θ on the support set of D_S of task T , which is calculated by Eq. (6), where f_{θ} denotes the model with parameters θ . Corresponding to the training error, $L(D_Q^T, \theta^T)$ is the test error of the model with parameters θ^T on the query set D_Q of task T , which is calculated by the following equation:

$$L(D_Q^T, \theta^T) = \sum_{(x,y) \in D_Q^T} \|f_{\theta^T}(x) - y\|_2^2 \quad (7)$$

where f_{θ^T} denotes the model with parameters θ^T . The test errors of all source tasks constitute the training error in the meta-training phase, and θ are updated by the following equation:

$$\theta = \theta - \beta \nabla_{\theta} \sum_T L(D_Q^T, \theta^T) \quad (8)$$

where β denotes the meta step size.

After the well-generalized initial parameters are acquired in the meta-training phase, the performance of the model is tested in the meta-testing phase. The model first calculates the training error on the support set D_S of the target task and updates the parameters to suit the target task by Eq. (5). The test error is then calculated on the query set of the target task to indicate model performance. The whole algorithm of MAML is outlined in Algorithm 1. After the process of updating parameters based on the training loss of the target task, the model will be more suitable for the data distribution of the target task.

3.3. Ensemble

The K subseries obtained by VMD have different frequencies and bandwidths, so K LSTM networks are trained separately to predict K subseries for more accurate prediction results, as shown in the “prediction” module of Fig. 1. After obtaining the prediction

Algorithm 1 MAML**Require:**

$\{T_{sou}\}$: the set of source tasks; $\{T_{tar}\}$: the set of target tasks;
 α, β : step size hyperparameters;
1: Randomly initialize θ
2: **while** not done **do**
3: sample batch of tasks $T \in \{T_{sou}\}$
4: **for all** T **do**
5: Calculate the training error $L(D_S^T, \theta)$ on the support set D_S
6: Update parameters by gradient descent: $\theta^T = \theta - \alpha \nabla_{\theta} L(D_S^T, \theta)$
7: Calculate the test error $L(D_Q^T, \theta^T)$ on the query set D_Q
8: **end for**
9: Update parameters by gradient descent: $\theta = \theta - \beta \nabla_{\theta} \sum_T L(D_Q^T, \theta^T)$
10: **end while**
11: **for all** $T \in \{T_{tar}\}$ **do**
12: Calculate the training error $L(D_S^T, \theta)$ on the support set D_S
13: Update parameters by gradient descent: $\theta^T = \theta - \alpha \nabla_{\theta} L(D_S^T, \theta)$
14: Make predictions on the query set D_Q
15: **end for**

Table 1

Stock symbols of stocks included in SSE and DJIA.

SSE	600000	600015	600016	600018	600028	600036
	600048	600050	600104	600111	600585	600637
	600795	600837	600887	600958	601006	601088
	601166	601169	601186	601211	601288	601318
	601328	601336	601398	601601	601628	601668
	601669	601688	601766	601800	601818	601857
	601901	601985	601988	601989	601998	
DJIA	AAPL	AMGN	AXP	BA	CAT	CRM
	CSCO	CVX	DIS	GS	HD	HON
	IBM	INTC	JNJ	JPM	KO	MCD
	MMM	MRK	MSFT	NKE	PG	TRV
	UNH	V	VZ	WBA	WMT	

results of K subseries, the following formula is used to obtain the predicted stock price:

$$Y_{t+1} = \sum_{k=1}^K \tilde{y}_{k(t+1)} \quad (9)$$

where Y_{t+1} denotes the predicted stock price at day $(t+1)$, and $\tilde{y}_{k(t+1)}$ denotes the model's prediction of the subseries u_{kt} .

4. Experiment

A series of experiments are conducted to validate the performance of VML. Dataset, parameter settings, and evaluation metrics of the experiment are first described in this section. Then, the experiments are briefly analyzed and discussed.

4.1. Experimental setups**4.1.1. Dataset**

To evaluate the prediction performance of the models in different markets, the experimental data are stock data of the Chinese market and the American market from January 2th 2018 to December 31th 2020 respectively. SSE dataset consists of stocks from the SSE 50 index of the Chinese market, and DJIA dataset consists of stocks from the Dow Jones Industrial Average of the American market. Stocks with missing price values in both

datasets are removed. The list of stocks for the two datasets is shown in Table 1. All data in the SSE dataset is obtained from Tushare, and that in the DJIA dataset is obtained from YahooFinance. For the stocks in both data sets, the daily closing price is used for the experiments. Firstly, the original stock price series of the two datasets are divided into training set, validation set and test set according to the ratio of 4:1:1. And next, the three sets are segmented with a sliding window and decomposed with VMD respectively. For decreasing the influence of noise, the samples are normalized according to the following equation:

$$\mu'_{kt} = \frac{\mu_{kt} - \min}{\max - \min} \quad (10)$$

where min and max denote the minimum and maximum values of the sample, respectively.

4.1.2. Parameter settings

We refer to the parameter settings of VMD in Niu et al. [20], the moderate bandwidth constraint is set to 2000, the noise tolerance is set to 0, and the tolerance of the convergence criterion is set to $1e-7$. The number of sample pairs for each task is set to 6, where 5 sample pairs constitute the support set and 1 sample pair constitutes the query set, i.e., D is 6, N_S is 5 and N_Q is 1. For LSTM, the hidden size is set to 8 and the learning rate is 0.01. For MAML, the learning rate is set to 0.01, and the number of gradient steps is 3.

The decomposition number K of VMD will affect the decomposition results and therefore the prediction results. Too little decomposition number K will lead to information loss or modal confusion, while too much decomposition number K will cause over-decomposition. In the same way as Jia et al. [44], we calculate the Pearson correlation coefficients of the residual and the original series to find the appropriate K . Jia et al. [44] proposed that when the original series is fully decomposed, the Pearson correlation coefficient of the residual and the original series should be minimal, corresponding to the optimal decomposition number K . When the decomposition number is K , the residual $\varepsilon_K(t)$ and Pearson correlation coefficients $P_K(f(t), \varepsilon_K(t))$ are calculated as:

$$\varepsilon_K(t) = f(t) - \sum_{k=1}^K u_k(t) \quad (11)$$

$$P_K(f(t), \varepsilon_K(t)) = \frac{\sum_{t=1}^N (f(t) - \bar{f})(\varepsilon_K(t) - \bar{\varepsilon}_K)}{\left[\sum_{t=1}^N (f(t) - \bar{f})^2 \sum_{t=1}^N (\varepsilon_K(t) - \bar{\varepsilon}_K)^2 \right]^{1/2}} \quad (12)$$

where $f(t)$ denotes the original series, and $u_k(t)$ denotes the IMF, N is the data length, $\bar{f}$ and $\bar{\varepsilon}_K$ denote the average of $f(t)$ and $u_k(t)$ respectively. We set K to 2, 3, 4, 5, 6, 7, 8, 9 respectively, and calculate the Pearson correlation coefficients. Fig. 5 shows the Pearson correlation coefficients for different K . From Fig. 5, it can be seen that the Pearson correlation coefficient decreases with an increase in K , but the change is small in the later stages. Furthermore, as K increases, the time required for decomposition increases. Therefore, we select an appropriate K by setting a threshold in order to balance the time efficiency and decomposition quality. Specifically, $k-1$ is chosen when $|P_K(f(t), \varepsilon_K(t)) - P_{K-1}(f(t), \varepsilon_{K-1}(t))|$ is less than the threshold (0.04).

To attenuate the effects of concept drift on the prediction, the time range where the samples are located cannot be too long, so the sliding window cannot be too large. The window size was set to 5, 10, 15, 20. In each fixed size window, the corresponding decomposition number K is determined according to the Pearson correlation coefficient. Table 2 shows the decomposition

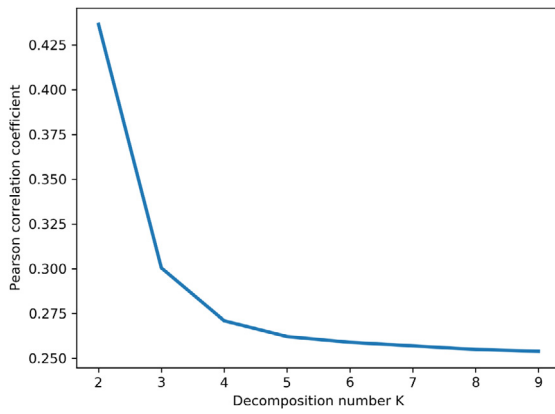


Fig. 5. Pearson correlation coefficients for different decomposition number K.

Table 2
Decomposition number K in different window sizes.

	5	10	15	20
SSE	3	3	4	4
DJIA	3	4	4	4

Table 3
Prediction performance in different window sizes.

		5	10	15	20
SSE	MSE	0.19	0.18	0.21	0.25
	MAE	0.18	0.18	0.20	0.22
	MAPE(%)	0.92	0.91	1.02	1.12
DJIA	MSE	7.49	7.05	8.82	9.41
	MAE	1.63	1.55	1.74	1.85
	MAPE(%)	1.11	1.05	1.18	1.26

number K for various window sizes on SSE and DJIA datasets. After determining the decomposition number, the experiments were conducted using windows of size 5, 10, 15, 20, which are shown in Table 3. According to Table 3, we can see that the best experimental results are obtained when the window size is 10.

4.1.3. Evaluation criteria

The commonly used mean squared error (MSE), mean absolute error (MAE) and mean absolute percentage error (MAPE) are adopted to evaluate the prediction accuracy of the model. They are respectively calculated as follows:

$$MSE = \frac{1}{N} \sum_{i=1}^N (X_i - Y_i)^2 \quad (13)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |X_i - Y_i| \quad (14)$$

$$MAPE = \frac{1}{N} \sum_{i=1}^N \left| \frac{X_i - Y_i}{X_i} \right| \times 100\% \quad (15)$$

where X_i is the actual stock price, and Y_i is the predicted stock price. N is the number of samples. MSE is the mean square error between the predicted and actual values, and MAE is the mean absolute error. MAPE is used to measure the relative error between the predicted value and the actual value, which is more suitable for comparison between data of different scales. Smaller MSE, MAE and MAPE values indicate higher prediction accuracy.

In addition, we evaluate the trading performance of the model using two statistical indicators Return and Sharpe Ratio. They are

Table 4

Prediction performance of three methods on SSE and DJIA.

		SFM [29]	PF-LSTM [45]	VML
SSE	MSE	0.25	0.27	0.18
	MAE	0.24	0.23	0.18
	MAPE(%)	1.31	1.18	0.91
DJIA	MSE	11.27	10.36	7.05
	MAE	2.05	1.96	1.55
	MAPE(%)	1.37	1.33	1.05

respectively calculated as follows:

$$R_p = \frac{X_p - X_0}{X_0} \quad (16)$$

$$\text{SharpeRatio} = \frac{E(R_p) - R_f}{\sigma_p} \quad (17)$$

In Eq. (16), X_0 denotes the stock price at the start of trading, X_p denotes the stock price at day p , and R_p denotes the Return at day p from the start of the transaction. In Eq. (17), $E(R_p)$ denotes the mean return, R_f denotes the risk-free rate, and σ_p denotes the standard deviation of the excess return. Generally, the higher Return and Sharpe Ratio, the better trading performance.

4.2. Prediction performance

To demonstrate the prediction performance, the VML method is compared with state-of-the-art methods, including State Frequency Memory(SFM) [29] and Particle Filter Recurrent Neural Networks (PF-RNN) [45]. SFM is proposed by Zhang et al. [29], which learns trading patterns with multiple frequencies to predict the trend of stock prices. Unlike RNNs that only focus on dependencies in the time domain, SFM pays attention to the information in both the time and frequency domain. PF-RNNs are proposed by Ma et al. [45] that combine the advantages of RNNs with particle filtering. PF-LSTM serves as a drop-in replacement for LSTM and is used in this paper.

Table 4 quantitatively shows the MSE, MAE and MAPE of three methods on the SSE and DJIA datasets. The values of MSE, MAE and MAPE of VML are the smallest for both datasets, which proves the effectiveness of the proposed VML model.

In Fig. 6, we plot the cumulative MAE curves of the SSE and DJIA datasets to demonstrate the performance clearly. In addition, the cumulative MAE curves of *China Petroleum & Chemical Corporation* (stock code: 601857) from the Chinese market and *Microsoft Corporation* (stock code: MSFT) from the American market are plotted separately. The horizontal scale represents the time in days, and the vertical scale reflects the cumulative MAE. The blue line shows the variation of the cumulative MAE of the PF-LSTM model, the orange line is the cumulative mean absolute error curve of the SFM model, and the red line represents the change of the cumulative MAE of the VML model proposed in this paper. As shown in Fig. 6, the curve of VML is always located at the bottom, indicating that VML has the smallest error. In addition, it is clear that the error gap between VML and the other two methods becomes larger and larger over time. Overall, these results indicate that SFM and PF-LSTM are subject to the concept drift. However, VML effectively mitigates the effect of concept drift and improves the prediction accuracy because of the continuous adaptation to the new data.

4.3. Ablation study

To prove the efficiency of the proposed method step by step, VML is compared with several methods: (1) LSTM; (2) MAML-LSTM, which combines LSTM with MAML; (3) VMD-LSTM, where

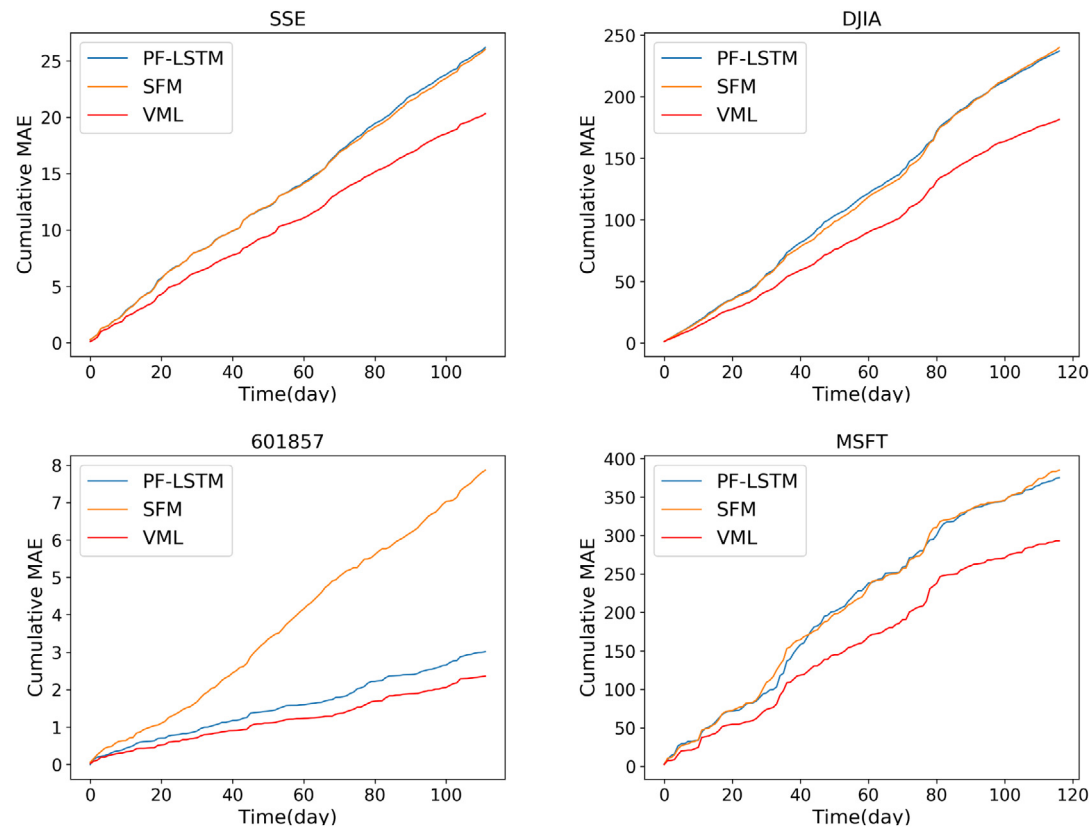


Fig. 6. Prediction performance of three methods.

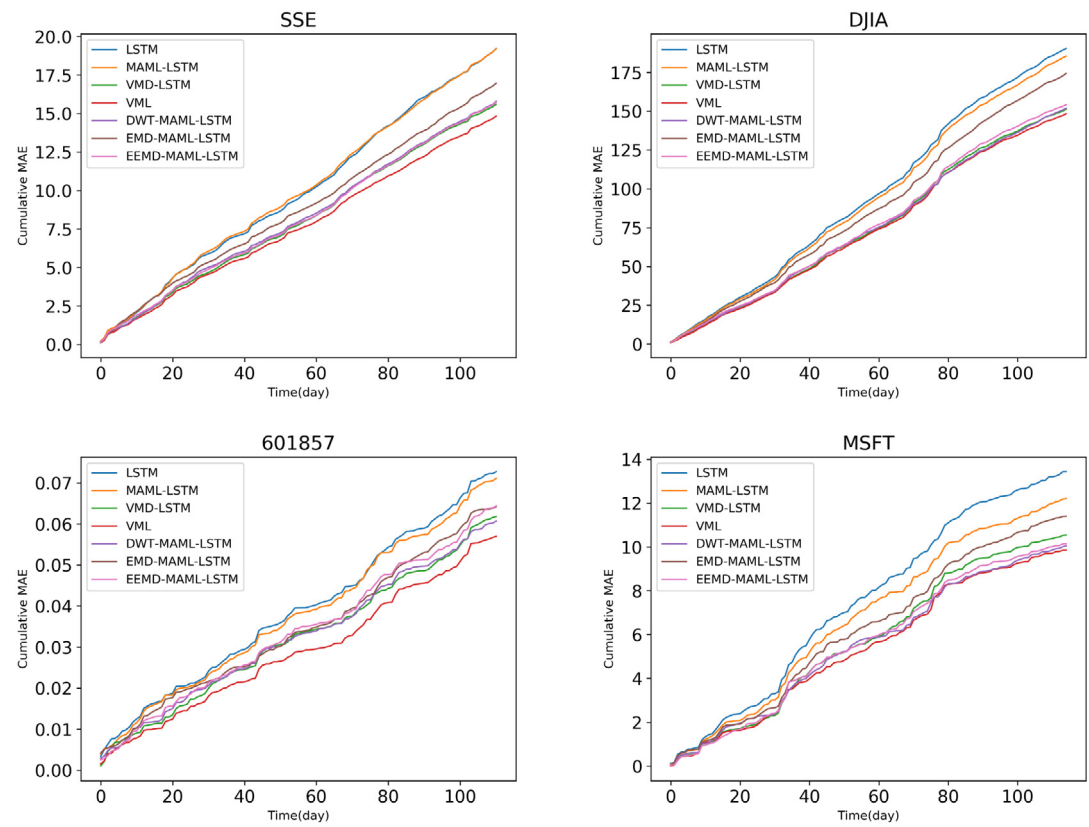


Fig. 7. Prediction performance of four models.

Table 5
Prediction performance on SSE and DJIA.

		SSE	DJIA
LSTM	MSE	0.29	11.34
	MAE	0.24	2.02
	MAPE(%)	1.20	1.36
MAML-LSTM	MSE	0.27	10.03
	MAE	0.23	1.93
	MAPE(%)	1.18	1.32
VMD-LSTM	MSE	0.20	7.24
	MAE	0.19	1.60
	MAPE(%)	0.96	1.08
DWT-MAML-LSTM	MSE	0.20	7.82
	MAE	0.19	1.77
	MAPE(%)	0.98	1.26
EMD-MAML-LSTM	MSE	0.24	7.54
	MAE	0.23	1.69
	MAPE(%)	1.04	1.20
EEMD-MAML-LSTM	MSE	0.18	7.63
	MAE	0.19	1.61
	MAPE(%)	0.95	1.09
VML	MSE	0.18	7.05
	MAE	0.18	1.55
	MAPE(%)	0.91	1.05

the decomposition object is changed from the whole series to window series; (4) DWT-MAML-LSTM, which replaces VMD of VML with DWT; (5) EMD-MAML-LSTM, which uses EMD to replace VMD of VML; (6) EEMD-MAML-LSTM, with EEMD to take the place of VMD.

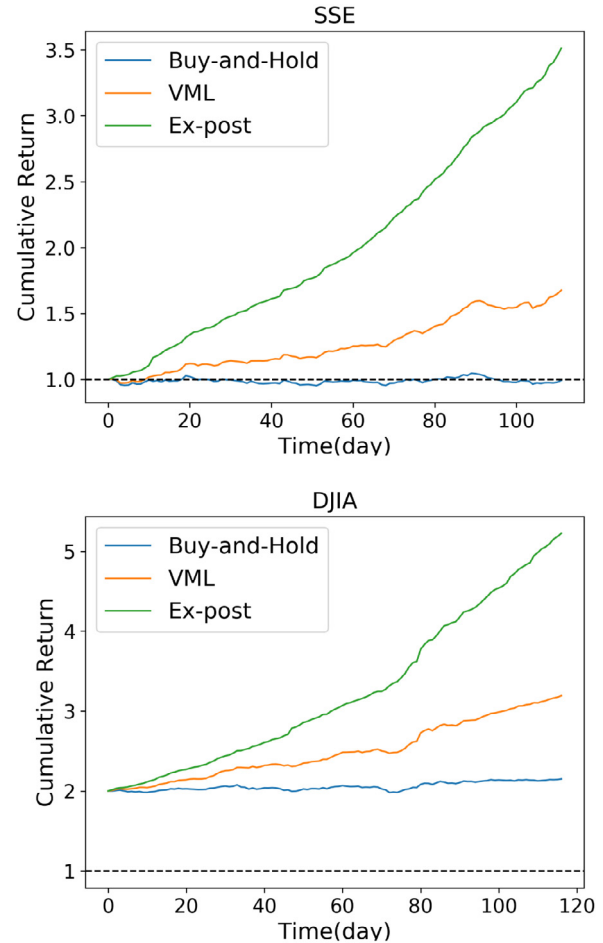
Table 5 quantitatively shows the MSE, MAE and MAPE of the seven models on the SSE and DJIA datasets. From the table, we can see that VMD-LSTM has smaller MSE, MAE and MAPE than LSTM. The same is true for VML compared with MAML-LSTM. The above results illustrate that these subseries can be modeled more effectively. MAML-LSTM has smaller MSE, MAE and MAPE than LSTM. The same is true for VML compared to VMD-LSTM. These results suggest that MAML can improve prediction performance. In addition, VML has better prediction performance than DWT-MAML-LSTM, EMD-MAML-LSTM, and EEMD-MAML-LSTM, indicating that VMD is superior to DWT, EMD and EEMD.

Fig. 7 shows the cumulative MAE curves of SSE and DJIA datasets and the cumulative MAE curves of *China Petroleum & Chemical Corporation* (stock code: 601857) and *Microsoft Corporation* (stock code: MSFT). The horizontal scale represents the time in days, and the vertical scale reflects the cumulative MAE. The blue, orange, and green lines indicate the cumulative MAE variations of LSTM, MAML-LSTM, and VMD-LSTM, respectively. The red line represents the change of the cumulative MAE of VML. The purple, brown and pink lines are the cumulative MAE curves of DWT-MAML-LSTM, EMD-MAML-LSTM, and EEMD-MAML-LSTM, respectively.

According to Fig. 7, we can draw the following conclusions: (1) the curves of VMD-LSTM and VML are below those of LSTM and MAML-LSTM, which shows the advantages of the decomposition-based hybrid model; (2) the curve of MAML-LSTM is below the LSTM curve, and the curve of VML is below the VMD-LSTM curve, illustrating that MAML is effective in boosting the model's performance; (3) the line of VML is always below the lines of DWT-MAML-LSTM, EMD-MAML-LSTM, and EEMD-MAML-LSTM, demonstrating that VMD outperforms DWT, EMD and EEMD; (4) the curve of VML is always at the bottom, indicating that VML has the highest prediction accuracy.

Table 6
Trading performance on SSE and DJIA.

		Buy-and-hold	VML	Ex-post
SSE	Return	0.993	1.677	3.513
	Sharpe	−0.18	7.29	27.8
DJIA	Return	1.151	2.192	4.223
	Sharpe	1.78	10.93	25.85

**Fig. 8.** Cumulative returns of three strategies on SSE and DJIA.

4.4. Trading performance

Higher prediction accuracy does not necessarily imply greater trading performance. We compute the profitability with a simple long-short trading strategy for the purpose of proving the reliability of VML in trading. The long-short trading strategy is specified as follows: if the price is predicted to rise, buy the stock and hold it until the stock price is predicted to fall. The trading performance of the long-short trading strategy is compared with that of the buy-and-hold and ex-post trading strategies. The buy-and-hold strategy implies buying a stock at the beginning and holding it until the end. The ex-post strategy is a look-ahead strategy assuming that the changes in stock prices are known. The ex-post strategy is an idealized strategy that cannot be implemented in practice but can provide an upper limit to the performance of the strategy.

When calculating the Return and Sharpe ratio, we assume that: (1) the market always has enough liquidity; (2) the buying and selling prices are the closing prices on the day of operation; (3) transaction cost is not considered. Table 6 shows the Return

Table 7
Experimental results of 1-step wind speed prediction.

		Inner Mongolia	Yunnan	Zhejiang
LSTM	MSE	0.69	0.83	0.77
	MAE	0.63	0.83	0.61
	MAPE(%)	13.49	13.32	14.24
ISSD-BP	MSE	0.49	0.61	0.52
	MAE	0.54	0.78	0.58
	MAPE(%)	12.79	12.68	13.74
ISSD-SVM	MSE	0.40	0.46	0.30
	MAE	0.46	0.68	0.46
	MAPE(%)	10.17	10.83	11.69
ISSD-LSTM	MSE	0.30	0.28	0.24
	MAE	0.45	0.57	0.49
	MAPE(%)	9.35	8.80	9.15
VML	MSE	0.24	0.26	0.34
	MAE	0.37	0.47	0.39
	MAPE(%)	8.79	8.47	8.45
VMD-LSTM-GOABDN	MSE	0.24	0.15	0.14
	MAE	0.35	0.43	0.36
	MAPE(%)	7.19	7.27	6.56
SSD-LSTM-GOABDN	MSE	0.21	0.10	0.08
	MAE	0.34	0.38	0.31
	MAPE(%)	7.22	5.83	5.01
ISSD-LSTM-GOABDN	MSE	0.08	0.05	0.03
	MAE	0.12	0.12	0.11
	MAPE(%)	2.56	2.63	2.06

and Sharpe ratio of three strategies. Since the future stock price is assumed to be known, the ex-post trading strategy has the highest Returns and Sharpe ratios on both datasets. In sharp contrast, the buy-hold strategy has the lowest Returns and Sharpe ratios. It is worth noting that the simple long-short strategy based on prediction results of the VML model performs well on both datasets, with significantly higher Returns and Sharpe ratios than the buy-hold strategy, indicating that the proposed VML has superior trading performance.

Fig. 8 presents the movement of the cumulative returns of the three strategies. The horizontal scale represents time in days and the vertical scale represents cumulative returns. The blue line indicates the cumulative return of the buy-and-hold strategy, which is at the bottom of both graphs. The orange line shows the change of the cumulative return of the long-short trading strategy determined by the forecast results of the VML. The green line represents the cumulative return of the ex-post strategy, which is at the top of both graphs. Affected by stock price fluctuations, the return curve of the buy-hold strategy has been oscillating without an upward trend. The curve of the ex-post strategy has been rising since the ex-post strategy can ideally avoid all stock price declines. The VML curve has oscillations, but compared with the buy-hold curve, the frequency of oscillations is significantly reduced. Moreover, the overall curve shows an obvious upward trend. In summary, these results show that VML can predict stock prices correctly, hence effectively avoiding investment risks and helping investors get favorable returns.

4.5. Comparison on wind speed prediction

There are various applications of time series forecasting. In addition to stock price prediction, we extend VML to wind speed prediction. Singular spectrum decomposition (SSD) is an adaptive method for decomposing nonlinear and non-stationary time series, proposed by Bonizzi et al. [46]. Yan et al. [12] improved SSD and used a hybrid model based on improved singular spectrum decomposition (ISSD) for wind speed prediction. In this study, experiments are conducted using the VML model for 1-step prediction of wind speed to compare with the ISSD-based

hybrid models. Wind speed data from paper [12] are used for the experiment.

The results are shown in Table 7. Based on Table 7, VML outperforms LSTM, ISSD-BP, ISSD-SVM and ISSD-LSTM in terms of prediction performance. However, VML has poor prediction capability in comparison to VMD-LSTM-GOABDN, SSD-LSTM-GOABDN and ISSD-LSTM-GOABDN. This is partly due to Yan et al.'s [12] use of two different models LSTM and DBN to predict the subseries. More importantly, these models decompose the original series and are prone to data leakage issues, leading to overly good prediction performance.

5. Conclusion

In this paper, a new stock price prediction method integrating meta-learning with the decomposition-based model, VML, is proposed. The decomposition object of the VML is changed from the whole series to window series, eliminating the look-ahead bias of the existing decomposition-based model. In addition, a meta-learning framework MAML is introduced to mitigate the impact of concept drift on stock price prediction. Experimental results indicate that the prediction performance of the VML is better than the state-of-the-art methods. Moreover, the back-testing results show that VML has excellent profitability. In summary, these results demonstrate the effectiveness of MAML and the superior performance of the decomposition-based model.

Further research may explore how to integrate the proposed model with quantitative systems for practical applications. Meanwhile, further investigation and experiments on the application of the proposed method in other scenarios, such as wind power forecasting, commodity demand forecasting, and traffic flow forecasting, are strongly recommended.

CRedit authorship contribution statement

Tengteng Liu: Conceptualization, Methodology, Software, Writing – original draft. **Xiang Ma:** Investigation, Data duration, Writing – review & editing. **Shuo Li:** Software, Data duration, Writing – review & editing. **Xuemei Li:** Supervision, Writing – review & editing. **Caiming Zhang:** Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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